

Subject Description Form

Subject Code	COMP 5422
Subject Title	Multimedia Computing, Systems and Applications
Credit Value	3
Level	5
Pre-requisite/Exclusion	Nil
Objectives	The objectives of this subject are to: 1. provide students with knowledge in fundamentals of multimedia, e.g. compression standards, data formats, media characteristics, storage and transmission requirements; 2. provide students with knowledge of a wide spectrum of multimedia information processing techniques; 3. train students with the ability to apply the knowledge in multimedia system and application development; 4. equip students with the ability to appreciate new and innovative solutions of multimedia systems and applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. have a good understanding on various characteristics of different media; 2. have a deep understanding of the techniques and requirements for processing multimedia, leveraging emerging technologies for global challenges; 3. grasp and consolidate the skills for problem solving involving multimedia databases, addressing complex interdisciplinary issues with professionalism; and 4. conduct real-world case study in multimedia applications, fostering intercultural and interdisciplinary interactions and demonstrating critical awareness and integrity in making responsible decisions.
Subject Synopsis/ Indicative Syllabus	• Multimedia System Primer: Introduction to different multimedia platforms, systems, tools and applications; characteristics of different media and current trends • Data Representation, Coding and Compression: Data representation, processing and analysis for Text, Sound/Audio, Image and Graphics, Video and Animation; Traditional coding requirements, compression techniques and standards: JPEG, MPEG, DVI, ASF, etc; Neural compression, codecs and large multimodal foundation models. • Multimedia Content Analysis and Information Retrieval: Multimedia contents: Color, shape, texture, motion, etc. • Traditional Content analysis techniques: Color histogram, shape analysis, motion analysis, etc. Multimodal knowledge extraction and integration. Multimedia retrieval techniques: object detection, scene understanding, video summarization, etc. Multimedia Security and Forensics: Traditional Digital watermarking, spatial and frequency domain techniques, livebitstream watermarking and content protection. AI-driven multimedia forensics: deepfake detection, adversarial attacks and defenses in multimedia systems.

	• Multimedia Information Networking: Video streaming, color models and motion estimation techniques, protocol support for multimedia networkings. AI-enhanced multimedia streaming systems and intelligent networking. • Selected Topics in Multimedia Computing, Systems and Applications: e.g., New MPEG standards, Multimedia Information Hiding and Watermarking, VoiceXML. Emerging applications and future directions for multimodal AI agents and embodied intelligence.																																	
Teaching/Learning Methodology	39 hours of class activities including - lecture, tutorial, lab, workshop seminar where applicable																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignments, Tests & Projects</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Final Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100</td><td colspan="4"></td></tr></table>						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Assignments, Tests & Projects	30	✓	✓	✓		2. Final Examination	70	✓	✓	✓	✓	Total	100				
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed																																
		a	b	c	d																													
1. Assignments, Tests & Projects	30	✓	✓	✓																														
2. Final Examination	70	✓	✓	✓	✓																													
Total	100																																	
Student Study Effort Expected	Class contact:																																	
	Class activities (lecture, tutorial, lab)		39 hours																															
	Other student study effort:																																	
	Assignments, Tests, Projects, Exams		66 hours																															
	Total student study effort		105 hours																															
Reading List and References	<i>Books</i> 1. Z.-N. Li, M.S. Drew, J. Liu, Fundamentals of Multimedia, Springer, 2004. 2. Lewis, Richard, 2005, Digital media: An introduction, Prentice Hall. 3. Borko Furht (ed), 1999, Handbook of Multimedia Computing. CRC Press. R. C. Gonzalez, Digital Image Processing, 4th ed., 2017 G. Friedland and R. Jain, Multimedia Computing, Cambridge University Press, 2014 <i>Journals</i> 1. IEEE Multimedia 2. IEEE Trans. on Multimedia 3. ACM SIG Multimedia																																	